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Palaeomagnetic timing of the rotation and translation of Cyprus

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The geological evolution of the Mesozoic Troodos Ophiolite Complex in Cyprus, and the tectonic nature and timing of the palaeomagnetically indicated anticlockwise rotation of Cyprus of some 80° and ca. 15° northward translation, have been open for debate for some time. New palaeomagnetic data from 18 sites (~ 180 samples) in the post-ophiolite sediments, ranging in age from Upper Cretaceous to Upper Miocene, are presented. Most of the sites are of normal geomagnetic polarity, but indications of reversed polarity have been found in an older group of sediments (the Lefkara Formation of Upper Palaeocene age).

Six sites from the older group of sediments (Upper Cretaceous to Eocene in age) give a site mean direction of the AF cleaned sediments of $(D, I) = (323°, 29°)$ with $\alpha_{95} = 18°$, while 5 sites from a younger group of sediments (Oligocene to Miocene in age) give a cleaned site mean direction of $(D, I) = (334°, 58°)$ with $\alpha_{95} = 9°$. These and published data suggest that an anticlockwise rotation of Cyprus of ~ $60 \pm 10°$ occurred early during the post-igneous evolution of the Cyprus oceanic crust between ~ 90 and ~ 50 Ma, leaving only a minor anticlockwise rotation of ~ $20 \pm 10°$ to occur during the last 50 Ma. It is furthermore concluded that the northward translation of Cyprus of ~ 15° mostly took place during the last ~ 30 Ma.

It thus appears that a fairly rapid rotation of the Cyprus microplate first took place in the Late Cretaceous and Early Tertiary time with an average angular velocity of ~ 1–2°/Ma, during which the northward translation was minor or negligible. In the latter half of the Tertiary, the sense of movement appears to have radically changed, the northward translation now being dominant with an average velocity of ~ 5–6 cm/yr. This temporal evolution is found to be in good agreement with the Mesozoic and Tertiary movements of the African lithospheric plate relative to Europe, as evidenced from the Atlantic sea-floor magnetic anomaly spreading history.

1. Introduction—previous palaeomagnetic work

The island of Cyprus in the eastern Mediterranean is a classical site for geological and geophysical interdisciplinary studies because of the presence of Cu-sulphide ore bodies, which have been mined far back in Antiquity [1]. In recent years the geology of Cyprus has been of special interest because of the Troodos Ophiolite Complex [2–6], which is believed to be part of an obducted slice of the oceanic crust, formed during the late Mesozoic.

The Tethys mobile belt of the Mediterranean area has been a collision zone from at least early Mesozoic time [9,10] to the present day [11,12] between the converging continents of Africa and Europe-Asia, and numerous palaeomagnetic investigations [9,13] suggest that several individual microplates suffered local rotations between the major continents, e.g. the Iberian Peninsula [14], Sardinia and Corsica [15,16], the Italian Peninsula [17–21], the Ionian Islands [22,23], and Turkey [24].

The Troodos Complex consists of ultramafics, gabbros, the sheeted diabase complex, and Lower and Upper pillow lavas, on top of which Late Cretaceous and Tertiary marine sediments were deposited, the sequence being elevated above sea level in the Late Tertiary [7]. The diabase forms a generally N-S trending dyke swarm, indicating considerable E-W extension and suggesting formation by sea-floor spreading [4]. Because of the exceptional opportunity to study an exposed section of former oceanic crust, detailed surface investigations in connection with the extensive drilling program of the ICRDG (International Crustal Research Drilling Group) have been undertaken since 1980 [8].

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Early palaeomagnetic work on the Troodos volcanics by Vine and Moores [3,4] suggested that since the Turonian Cyprus rotated some 80° anticlockwise and migrated northwards some 16° to the present latitude of 35°N. Although the magnetic sampling was elaborate, no consistent reversals were found [25–27], and it was concluded that the whole complex was probably of Middle Cretaceous age, consistent with the Turonian age (~ 91–88 Ma) assigned to the overlying radiolarian shales [28]. Anomalous magnetizations in pillow lavas probably indicative of a reversal were later reported by Schönharting and Abrahamsen [29], while independent palaeomagnetic results from the younger sediments supported the anticlockwise rotation of Cyprus, although the timing of the rotation was then left open [30].

Palaeomagnetic results from the sediments led Shelton and Gass [31] to suggest that "rotation took place only in the geologically recent past", i.e. within the last ca. 12 Ma, whereas Clube et al. [32,33] from a new set of palaeomagnetic data found that "a major component of the ophiolite rotation occurred within the Late Cretaceous–Early Tertiary interval rather than the Miocene".

The purpose of the present paper is to present our palaeomagnetic results from the younger sediments in more detail than hitherto [30], and to discuss the timing of the possible rotation and translation of the Cyprus microplate. As to the question of the time of the rotation our conclusion is in accordance with that of Clube et al. [32,33,44].

2. Magnetic methods and results

The palaeomagnetic field work was done in May 1980, when 18 sites (~ 180 samples) in the post-ophiolite sediments were sampled (Fig. 1) ranging in age from the Umber of Upper Cretaceous age to the Koronia Formation of Upper Miocene age (Table 1). Some of the samples of the well consolidated chalks of the Lefkara Formation were cored in the field (sites L4 to L6) by a water-cooled portable drill, but the majority of the sediment sites were sampled by means of oriented hand samples. In all cases the orientations were made by sun compass, and checked by magnetic compass. At each site a set of at least 10 specimens was aimed at, although not all of them survived the succeeding preparations in the laboratory. The strike and dip of the bedding was measured by an averaging procedure involving at least 10 individual determinations at each site, to make tectonic corrections of the site mean remanent directions possible (Table 1). In the laboratory, standard palaeomagnetic 25 mm by 22 mm long cylindrical specimens were finally prepared.

The remanent magnetization of most specimens was measured on a three-axis Squid magnetometer, alternating with successive step-wise partial demagnetizations in alternating peak magnetic fields up to 600 Oe (60 mT); some typical results are illustrated in the stereograms of Fig. 2. The site mean intensity of the NRM was quite low, between 0.1 and 0.5×10^{-3} A m^{-1} in 10 out of 15

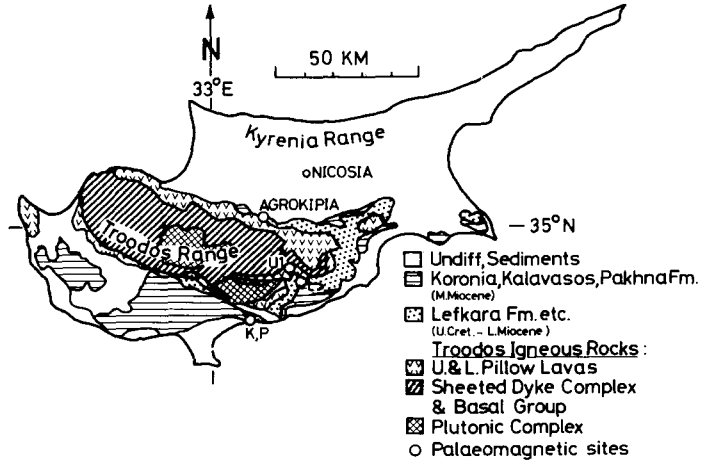


Fig. 1. Geological map of Cyprus, with palaeomagnetic sites shown (simplified from [40]).

TABLE 1

Cyprus sediments: cleaned palaeomagnetic mean directions [30]

Site		Age	Bedding (°)		N_1	N	J_0 (μG)	$\bar{F}$ (Oe)	$\bar{D}$ (°)	$\bar{I}$ (°)	α_{95} (°)	k	R	Tect. corrected	
			strike	dip										D_T (°)	I_T (°)
U1	Umber (north of Lefkara)	U. Cretaceous	35	17	7	7	5.2	100–400	305	36	11	30	6.7998	305	53
L1	Lefkara Fm. (Lefkara)	U. Palaeocene	65	$4\frac{1}{2}$	8	8	0.3	100–400	123	48	13	20	7.6424	126	44
L3a		U. Palaeocene	95	$5\frac{1}{2}$	6	6	0.5	100–400	162	30	40	4	4.6894	163	25
L3b		U. Palaeocene	100	12	4	4	0.4	400	110	−33	56	4	3.1894	102	−34
L4		Eocene		0	11	11	1.1	100–400	321	19	6	65	10.8458	321	19
L5		L. Eocene	41	11	6	6	0.8	0–200	345	29	8	64	5.9221	349	38
L6		L. Eocene	35	14	7	7	0.5	1–400	344	27	10	37	6.8391	350	37
A0	(Akrokipia)	Eocene (below zone 1 [a])	290	28	8	8	16	0–200	335	18	21	8	7.0896	338	−2
A1		?U. Oligocene (zone 1)	295	28	6	6	0.3	0–200	326	53	16	20	5.7438	346	34
A2		L. Miocene (zone 2)	290	28	9	9	0.2	0–600	323	56	13	16	8.4935	344	36
A3		L.m. Miocene (zone 3)	290	28	6	5	1.1	0–600	354	61	13	37	4.8905	5	35
A5		M.m. Miocene (zone 5)	290	28	7	7	0.3	100–200	351	63	10	39	6.8443	4	37
A7		U.m. Miocene (zone 7)			6		0.03								
K	Koronia Fm. (Amathus)	M. to U. Miocene	80	10	12	11	0.5	100–400	326	52	20	6	9.6437	319	61
P	Pakna Gr. (Amathus)	M. Miocene	90	5	9		0.2								
Mean of sites															
Older group:	sites U1, – L3b, L4, L5, L6:					5			322	31	39	3.9	4.7160		
											26	9.9	4.5948	322	39
	sites U1, – L3b, L4, L5, L6, A0:					6			323	29	18	14.8	5.6625		
											26	7.8	5.3628	326	32
Younger group:	sites A0, A1, A2, A3, A5:					5			336	51	19	17	4.7547		
											19	17	4.7612	351	29
	sites A1, A2, A3, A5, K:					4			337	59	11	71	3.9558		
											11	77	3.9608	354	36
	sites A1, A2, A3, A5, K:					5			334	58	9	76	4.9473		
											16	23	4.8246	350	42
All sites (except L1 and L3a):						11			328	42	13	13.6	10.2626		
											15	10.1	10.0069	336	37

[a] Foraminiferal zones [41].

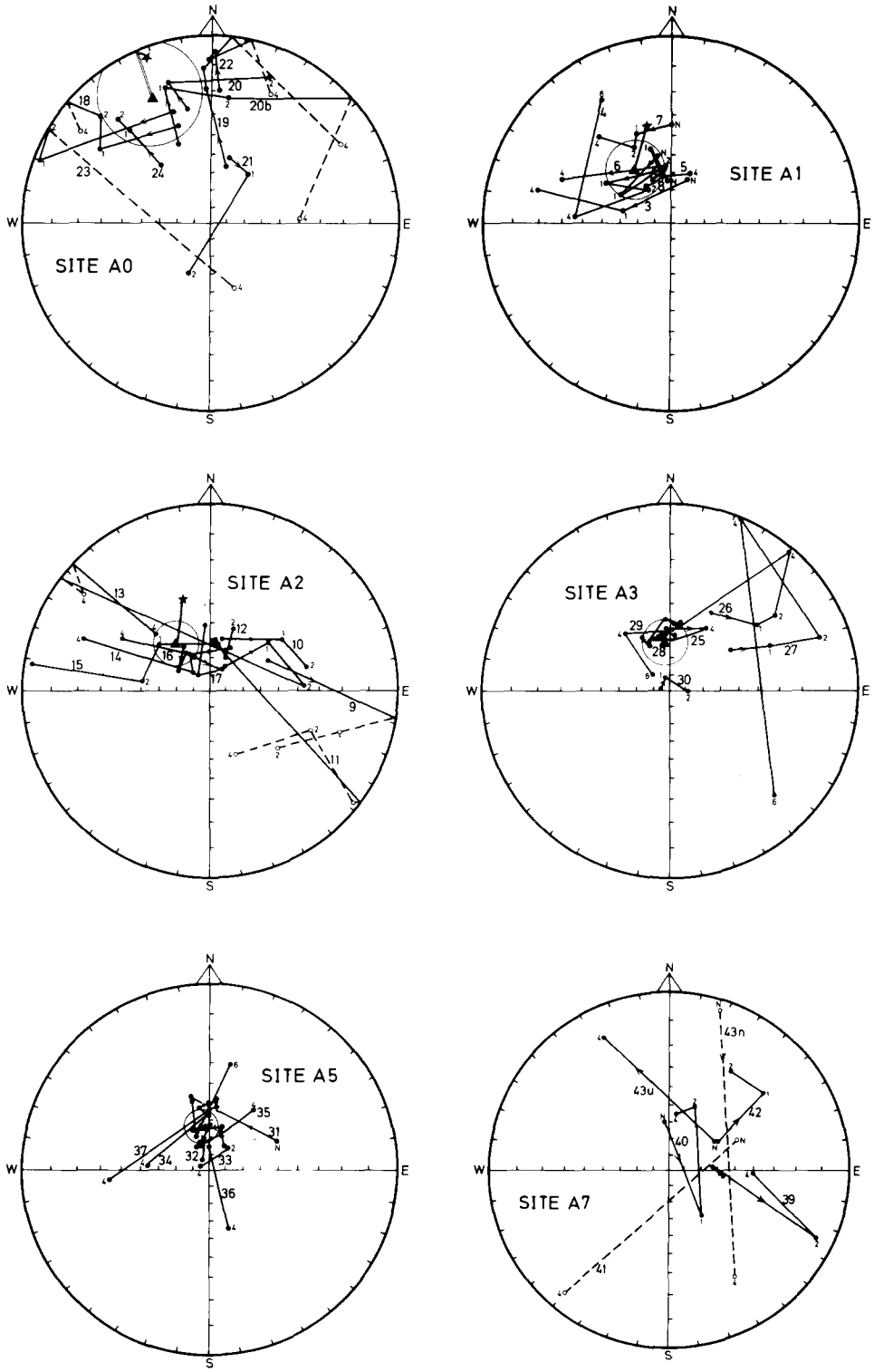


Fig. 2. Agrokipia sites A0 to A7, Eocene to Upper middle Miocene. AF demagnetization of individual specimens. Solid (open) symbols indicate positive (negative) inclinations without tectonic corrections. Circles show α_{95} (95% significance) cones of error around the site mean (triangle is uncorrected, and star is tectonically corrected mean). Stereographic projection.

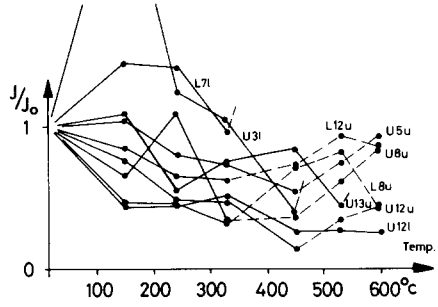


Fig. 3. Decay of intensity of the remanent magnetization by thermal demagnetization of specimens from sites K and P.

sites, and the median destructive field (MDF) was typically between 50 and 100 Oe (5–10 mT).

Furthermore, partial thermal demagnetizations and repeated measurements by means of a spinner magnetometer on 11 specimens gave somewhat directionally scattered results, which may be ascribed partly to the low intensities ($\sim 10^{-4}$ A m^{-1}) and partly to oxidizing and remagnetization processes in the oven during cooling. Although the decrease in J/J_0 by increasing temperature is somewhat scattered too (Fig. 3), Curie points in the range 350–450°C are likely to be present.

In combination with the low values of the MDF this may indicate that the dominant, carriers of the remanence are polydomain detrital titanomagnetites or titanomaghemites of igneous origin derived either from the Troodos Complex (rather than Ti-containing magnetites of continental origin) or from magnetites formed by diagenetic processes in the sediment. However, no ore microscopic investigations have yet been conducted, and the possibility that the "Curie temperatures" are in fact temperatures at which the magnetic phase decomposes into non-magnetic phases cannot be excluded.

The cleaned directions of site L1 and L3a are in a southeasterly direction, but with positive inclinations. We suggest that they may originate from a transition between two polarities, or they could be the effects of a partial remagnetization in a reversed polarity field. Because of the anomalous directions, the two sites are discarded from further discussions.

The Fisher statistics in Table 1 are somewhat improved by tectonic correction in the case of the mean of the 5 older sites (Umber and Lefkara Formation), the precision parameter being increased from $k = 3.9$ to 9.9 ($N = 5$). If, however, site A0 (Agrokipia, Eocene) is included in the mean of this older group, the precision is decreased from $k = 14.8$ to 7.8 by tectonic correction. This is also the case with site K, if this site is included in the mean of the younger group, reducing the precision of this group from $k = 76$ to 23 with tectonic correction. This suggests that sites A0 and K have been partly remagnetized after the tectonic tilting.

Although the difference is minor, we conclude that in most cases (except site A0 and K) the cleaned remanent magnetization was locked in before the tectonic tilting occurred, whence the remanence should be tectonically corrected before being used in tectonic considerations (Table 1). However, our final conclusions are unaffected whether or not one applies tectonic corrections to our data, since the differences are insignificant.

Because of the variable quality due to the varying scatter of the sedimentary palaeomagnetic data, we divide the following discussion into two parts, related to the changes in inclination (indicative of a northward translation) and in declination (indicative of rotation), respectively.

3. Palaeomagnetic evidence for the timing of the northward translation

Although an anticlockwise rotation of Cyprus of ca. 80° and a northward translation of some 15° since the formation of the Troodos Complex (not younger than 90 Ma) were reported by Vine and Moores in 1969 [3], and more fully discussed in their conclusions based on palaeomagnetic data from the igneous rocks [4], the more detailed timing of the rotation and translation has been difficult to determine. This is because the undeformed younger sediments of post-Troodos age, which must carry the relevant palaeomagnetic information, are generally very weakly magnetic, being dominantly calcareous. Hence the directional scatter of the remanence is high even when measured on a low-noise spinner magnetometer. This was demonstrated by the preliminary, although quite detailed, work of Shelton and Gass (table 1 in [31]), who in 9 out of 12 sediment sites found $\alpha_{95} > 40°$, and in the remaining 3 sites found $\alpha_{95} \sim 25°$.

On a Squid magnetometer the noise level is typically one order of magnitude lower. Thus our data (Table 1) show less directional scatter, with α_{95} typically between 6° and 20° (11 sites out of 14), although the scatter of individual specimens was still high, as can be seen in Fig. 2.

In Figs. 4 and 5 the AF cleaned site means (after applying tectonic corrections) of the older and younger group of sediment sites, respectively, are shown, according to Table 1.

Most of the sites are normally magnetized to the north or northwest, but one site, L3b (Lefkara Formation of Upper Palaeocene age) is reversely magnetized, although the scatter is high. The bedding corrected site mean direction of the older group of sediment sites (Umber and Lefkara Formation, Upper Cretaceous to Eocene) is (D_T, I_T, N, α_{95}) = (326°, 32°, 6, 26°). Without bedding correction the mean direction is practically unchanged ($\bar{D}$, $\bar{I}$) = (323°, 29°), while the scatter is somewhat less, with $\alpha_{95} = 18°$, as discussed in the previous section.

The younger group of sediment sites (A1 to A5 and K, of Oligocene to Miocene ages) have a

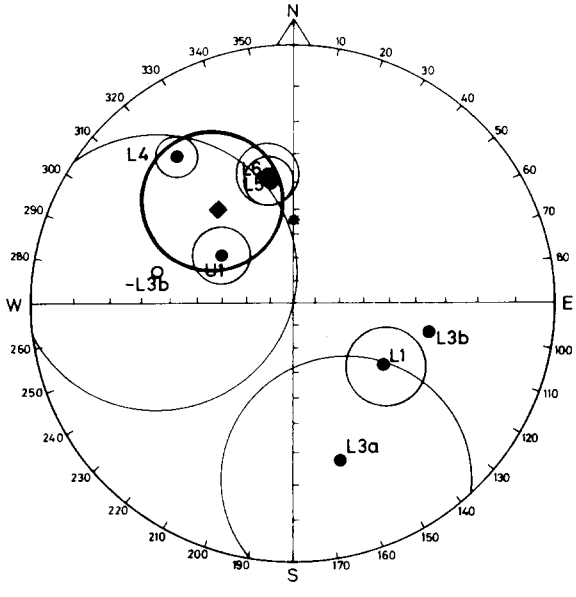


Fig. 4. Site means after AF cleaning and tectonic corrections (cf. Table 1) of the older group of sediment sites, with α_{95} circles. Site L3b has been reversed and used in the mean shown (diamond and fat circles). Asterisk indicates the present geomagnetic field. Stereographic projection.

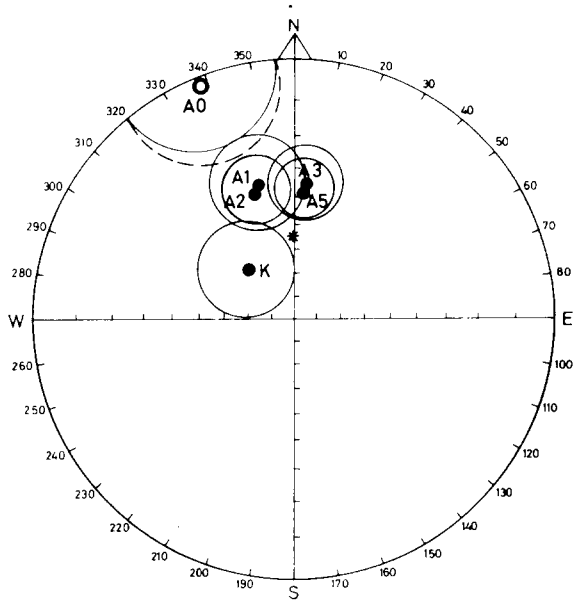


Fig. 5. Site means after AF cleaning and tectonic corrections of the younger group of sediment sites (cf. Table 1).

tectonically corrected mean direction of (D_T, I_T, N, α_{95}) = (350°, 42°, 5, 16°), while the mean direction without tectonic correction is (334°, 58°, 5, 9°). The tectonically corrected low values of inclinations for both age groups of sediment sites (32–42°) and the overall mean of 37° (equivalating a mean palaeo latitude of $\phi \simeq 21°$ N) compare well with the low inclination value of 34° (equivalent to a palaeolatitude of $\phi \simeq 19°$ N) found by Vine and Moores [3] for the Troodos Complex, and is in contrast to the present dipole inclination of 54.5° (present latitude of 35° N).

In Fig. 6 the inclinations from Table 1 are plotted versus age, together with data from Shelton and Gass [31] and Moores and Vine [4]. We may now set up and test two hypotheses regarding the change in inclination with time. One hypothesis is that the inclination has changed linearly with time since 90 Ma until now. The other hypothesis is that the major change (i.e. northward drift) occurred during the last ca. 30 Ma, that is since Upper Oligocene. It is obvious from the figure that the latter hypothesis fits the data better than the former.

Although the inclinations are somewhat scattered we therefore conclude that the magnetic

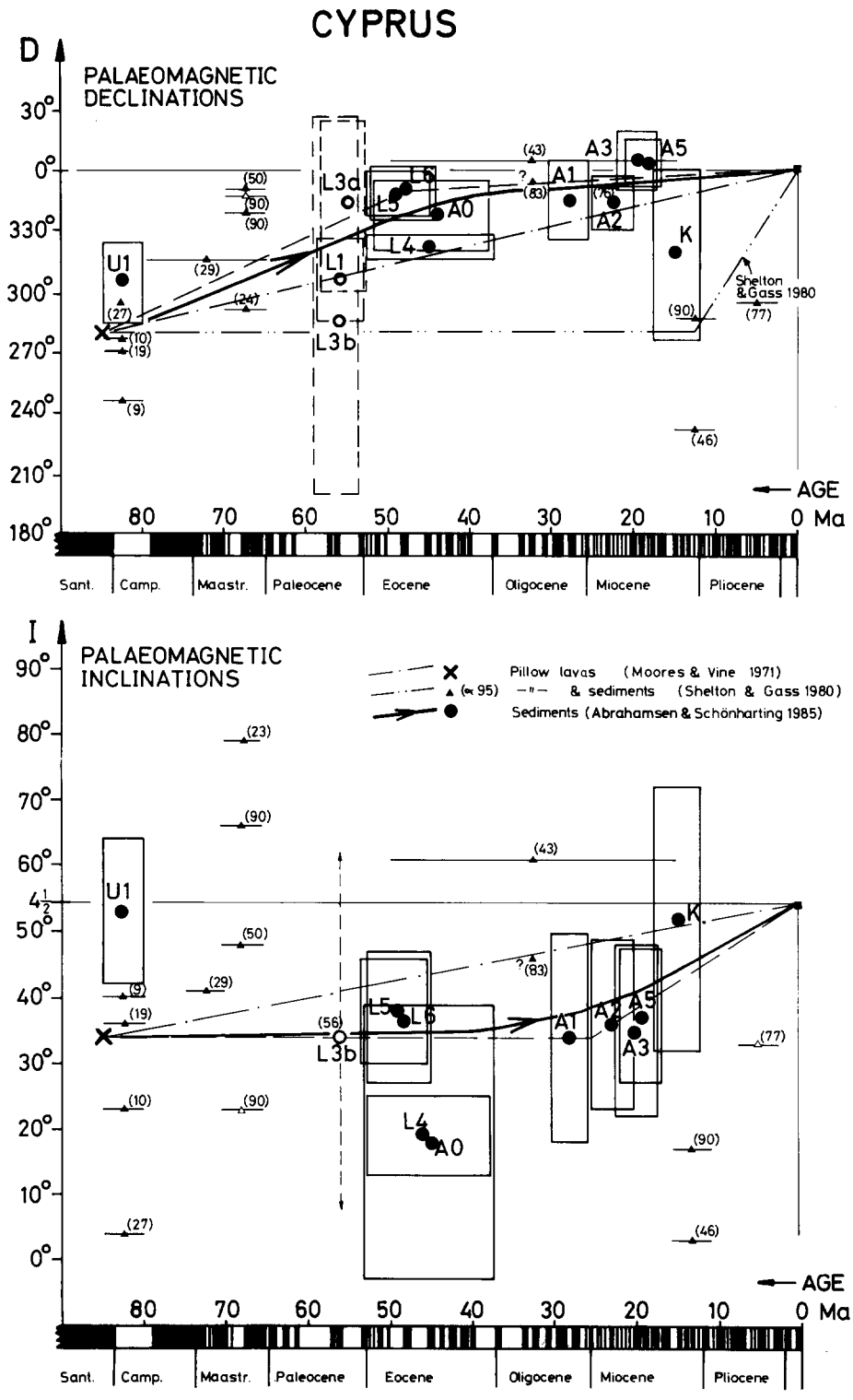


Fig. 6. Compilation of declinations and inclinations versus age for sediments from Cyprus. Error boxes indicate the full 95% significance angle (for the declination $\alpha_{95}/\cos I$), while small numbers in brackets indicate α_{95} only. Polarity scale is from Cox [26].

data from sediments indeed support the result by Vine and Moores [3,4] that Cyprus was translated ca. 15° towards the north; it further appears that most of this northward drift took place during the last ca. 30 Ma, as indicated by the curved line in Fig. 6.

4. Palaeomagnetic evidence for the timing of the anticlockwise rotation

The palaeomagnetic declinations from Table 1 are also plotted versus age in Fig. 6, together with the data from Shelton and Gass [31] and Moores and Vine [4].

The previous conclusion of Moores and Vine and of Shelton and Gass that Cyprus has significantly rotated anticlockwise since the formation of the Troodos Complex is also supported by our sediment data, although we leave open the question of the exact amount of the rotation, as we have added new data of one site only from the oldest part of the sediments (site U1).

Shelton and Gass [31] found it probable that the rotation took place in the last 10 Ma. Their data (table in [31]) from the site of Pendaskinos (No. 5035) and the site of Tsadha (No. 5044) both of Miocene age, however, deserve some discussion, as these sites may tend to be the ones which mainly supported their conclusion of a late rotation. The large values of the confidence angle α_{95} of 45.9° at site 5035 and of 90° at site 5044 as well as the fact that the data represent uncleaned directions, decrease the importance of these data considerably. In addition, low inclination values of 2.7° and 16.9°, respectively, may be indicative of mixed polarities being responsible for the scatter. The possibility of either true reversals and/or later remagnetization may be tested in the future by resampling these sites in detail.

From our data, as well as from the other data published [3,4,30–33], we find it more likely, however, that the major part of the rotation (say $60 \pm 10°$) took place in the early part of the time interval in question between ca. 90 and ca. 50 Ma, as suggested by the full curve in Fig. 6, leaving only a minor rotation of say $20 \pm 10°$ to occur during the last 50 Ma.

It thus appears that a fairly rapid rotation of the Cyprus microplate first took place in the late Cretaceous and early Tertiary with a mean velocity of rotation of the order of 1–2°/Ma, during which time the northward translation was minor or negligible. In the younger half of the Tertiary the sense of movement then appears to have changed, the northward translation being dominant with an average of ca. $\frac{1}{2}°$ per Ma or ~ 5–6 cm/yr northwards, while the rotation nearly (or fully?) stopped with a maximum average velocity of rotation of only ~ $\frac{1}{2}°$ per Ma or less.

5. Tectonic implications for the timing of the rotation and translation of Cyprus

The palaeomagnetically suggested sequence of rotation and translation of Cyprus gives some new constraints for plate tectonic models in the eastern Mediterranean. The early timing of the main component of rotation as opposed to the conclusions of Shelton and Gass [31] helps in the understanding of the facts that

(1) no oceanic crust has been formed during the last 10 m.y. around Cyprus,

(2) no major deformation zones have been formed during this period on or around Cyprus apart from the arcuate shaped active microseismic zone south of Cyprus (Neophytou, personal communication 1981), and

(3) the heat flow of Cyprus and generally of the region around Cyprus is very low [34].

In suggesting a plate tectonic or microplate tectonic setting for Cyprus, the major structural, petrological and geochemical findings on Troodos igneous rocks have to be taken into account. The discussion on the significance of geochemical trends continues, and various new tectonic models have been presented recently (e.g. [35,36]) placing the Troodos oceanic crust in a subduction zone environment with fore arc, island arc, back arc, or marginal seas characteristics.

Palaeomagnetic results from Cyprus alone are not sufficient to choose between these models. In the following only a few aspects are considered which are related to the tectonic evolution of the area.

Accepting early rotation we suggest that the Troodos oceanic crust collided with surrounding blocks soon after formation. This can be accomplished with a subduction zone fairly close by. The model by Moores et al. [36] which compares the Cyprus and Arabian block ophiolites with the

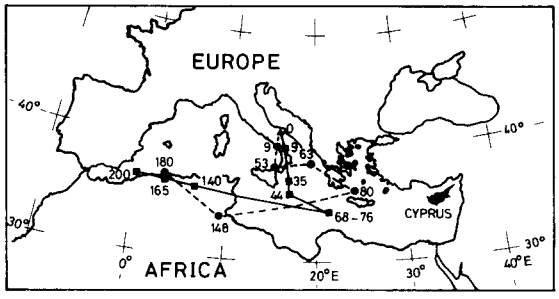


Fig. 7. Movement of Africa relative to Europe since the Early Mesozoic as derived from Atlantic magnetic anomaly data. Dotted line: according to Pitman and Talwani [37] and Dewey et al. [38]. Full line: modification made by Biju-Duval et al. [39]. Numbers are ages in Ma (redrawn from Channell et al. [9]). More recent analyses [43] of the Atlantic spreading data confirm the dominant easterly translation of Africa relative to Europe in the time interval 190 to 80 Ma, and a dominant northerly convergence between 80 and 0 Ma, whereas the intermediate western translation shown above is not confirmed.

present situation of the Andaman Sea, may represent a good case for the geodynamic setting of Troodos, however, without their interpretation of a late, substantial rotation of Cyprus.

The proposed early rotation of Troodos separates in time the main rotation component and the time of emplacement and makes them two definitely distinct events. The collision of a subduction zone with the African-Arabian plate, as suggested to be the mechanism of obduction by Moores et al. [36], however, may still be a viable mechanism for obduction and for the second minor and later component of rotation.

In addition to these elements a regional shear-strain resulting from the eastward movement of the African plate relative to Europe until about 80 to 60 m.y. ago, as deduced from the Atlantic marine magnetic anomalies (Fig. 7, from Channell et al. [9]), may contribute also to an anticlockwise rotation component of the Troodos rotation during the early stages, provided that Troodos was situated close to the belts of major left lateral displacements.

Finally, the palaeomagnetic timing of the northward movement of Cyprus to the younger Tertiary finds its counterpart also in Fig. 7, where the African plate is shown to be migrating northward relative to Europe in Tertiary time. Returning to the palaeomagnetic information in Fig. 6 we therefore suggest that Cyprus was obducted onto the African plate some time between 50 and 30 m.y. ago, and since then is to be regarded as a part of the African plate, moving northward towards the European and Anatolian plates [42].

Future palaeomagnetic sampling from eastern Mediterranean areas around Cyprus may add to constrain the possible plate tectonic solutions for the area.

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